Fourier Lemmas on the Three-Torus: Heat Smoothing, the L^2 Strain Identity, and the Curl of the Energy Derivative, Formalised in Lean 4

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Abstract

On $\mathbb{T}^3 = (\mathbb{R}/\mathbb{Z})^3$, the product of three circles, we prove in Lean 4 with Mathlib, with no `sorry` and only the standard axioms: the heat semigroup is the Fourier multiplier $e^{-\nu|n|^2 t}$, obeys the semigroup law and contracts L^2; it maps H^s to H^{s+k} with $\|e^{t\nu\Delta} f\|^2_{H^{s+k}} \le k!\, e^{2\nu t}(2\nu t)^{-k}\|f\|^2_{H^s}$; for divergence-free velocity fields $\|S\|_2^2 = \frac{1}{2}\|\omega\|_2^2$; and the kinetic energy, viewed as a function of vorticity, has derivative the stream function, whose curl is the velocity. For the finite-mode Galerkin truncation we further prove local existence, preservation of the Hermitian and divergence-free structure, an H^s transport bound for $s > 5/2$, a uniform-in-N short-time bound, a tame Grönwall propagation estimate, and continuation under an a priori bound. The full build (14 modules, 250 theorems) uses no `sorry` and only the standard axioms. These are Galerkin (finite-mode) and Fourier statements; nothing here proves existence or regularity of Navier–Stokes solutions, and the Clay Millennium problem remains open.

1 Setting

$\mathbb{T}^3$ carries normalised Haar measure; $e_n(x) = e^{2\pi i n\cdot x}$, $n \in \mathbb{Z}^3$, is a Hilbert basis of $L^2(\mathbb{T}^3)$ (Mathlib's `mFourierBasis`), and $\hat{f}(n) = \int f\,\overline{e_n}$.

Lemma 1 (Parseval, `parseval_norm_sq`)**.** $\|f\|_2^2 = \sum_n |\hat{f}(n)|^2$.

Proof. The coefficient map is a linear isometry $L^2 \to \ell^2(\mathbb{Z}^3)$ and $\|a\|^2_{\ell^2} = \sum |a_n|^2$. □

Lemma 2 (`norm_multiplier_le`)**.** *If $|m(n)| \le 1$ for all n, then $T_m f = \sum m(n)\hat{f}(n)e_n$ defines $T_m : L^2 \to L^2$ with $\|T_m f\|_2 \le \|f\|_2$.*

Proof. $\sum |m(n)\hat{f}(n)|^2 \le \sum |\hat{f}(n)|^2 < \infty$, so the sequence lies in ℓ^2; pull back by the isometry and apply Parseval. □

Commentary. The earlier manuscript already ran the analysis through Fourier coefficients: "The Fourier multiplier proof of the Sobolev embedding $H^s(\mathbb{T}^3, \mathbb{R}^3) \hookrightarrow C^1(\mathbb{T}^3, \mathbb{R}^3)$ for $s > 5/2$ follows from Parseval's identity and the p-series convergence" (*vFinal*, author's unpublished draft). When this build was planned, the author's point was simply: "the torus is proved to be a Circle (cubic)" (30 October 2026). $\mathbb{T}^3$ is three circles, so Mathlib's multidimensional Fourier series and Parseval apply directly, and the Sobolev scale, missing from Mathlib, is defined here as a weighted coefficient sum.

2 Heat semigroup

Theorem 3. *For $\nu, s, t \ge 0$: $\widehat{e^{t\nu\Delta} f}(n) = e^{-\nu|n|^2 t}\hat{f}(n)$, $e^0 = \mathrm{id}$, $e^{s\nu\Delta}e^{t\nu\Delta} = e^{(s+t)\nu\Delta}$, $\|e^{t\nu\Delta} f\|_2 \le \|f\|_2$.*

Proof. The symbol has modulus ≤ 1; $e^{-\nu|n|^2 s}e^{-\nu|n|^2 t} = e^{-\nu|n|^2(s+t)}$ and operators with equal coefficients are equal. □

Commentary. The local-existence block of the May drafts used exactly this operator: "The heat semigroup $e^{\nu t\Delta}$ is bounded on H^s ($s > 5/2$) with $\|e^{\nu t\Delta}\|_{\mathcal{L}(H^s)} \le 1$" (*May 16 edits*, author's unpublished draft). There it was asserted; here it is the contraction theorem above, for every s, since the symbol has modulus at most one at every mode.

3 Sobolev smoothing

$\|f\|_{H^s}^2 = \sum_n (1 + |n|^2)^s |\hat{f}(n)|^2$.

Lemma 4. *For $k \in \mathbb{N}$, $a > 0$, $x \ge 0$: $(1 + x)^k e^{-ax} \le k!\, e^a a^{-k}$.*

Proof. $y^k/k! \le e^y$ with $y = a(1 + x)$, and $e^{-ax} = e^a e^{-a(1+x)}$. □

Theorem 5 (`heat_smoothing`)**.** *For $\nu, t > 0$, $s \in \mathbb{R}$, $k \in \mathbb{N}$, $f \in H^s$: $e^{t\nu\Delta} f \in H^{s+k}$ and $\|e^{t\nu\Delta} f\|_{H^{s+k}}^2 \le k!\, e^{2\nu t}(2\nu t)^{-k}\|f\|_{H^s}^2$.*

Proof. Mode by mode, $(1+|n|^2)^{s+k} e^{-2\nu t|n|^2}|\hat{f}|^2 \le k! e^{2\nu t}(2\nu t)^{-k}(1+|n|^2)^s|\hat{f}|^2$ by the lemma with $a = 2\nu t$; sum by comparison. □

Theorem 6 (`heat_smoothing_real`)**.** *The same holds for every real $k \geq 0$ with $k!$ replaced by $\lceil k \rceil !$.*

Proof. For $y \geq 0$, $y^k \leq \lceil k \rceil ! \, e^y$: if $y \leq 1$ then $y^k \leq 1 \leq e^y$; if $y \geq 1$ then $y^k \leq y^{\lceil k \rceil} \leq \lceil k \rceil ! \, e^y$. The rest is as before. □

Commentary. The v2.4 master draft names this step as indispensable: "Viscosity is essential for local existence in H^s, the explicit higher-norm Gronwall estimate, and the final parabolic smoothing" (*NS Global Regularity v2.4 Master*, author's unpublished draft, 25 May 2026). The drafts invoked the smoothing without a statement; the theorem above gives it with an explicit constant.

4 Strain and vorticity

For $k \in \mathbb{R}^3$, $\hat{u} \in \mathbb{C}^3$, $k \cdot \hat{u} = 0$: $\hat{S}_{ij} = \pi i (k_i \hat{u}_j + k_j \hat{u}_i)$, $\hat{\omega} = 2\pi i \, k \times \hat{u}$.

Lemma 7. $\sum_{ij} |\hat{S}_{ij}|^2 = \frac{1}{2} |\hat{\omega}|^2$ *and* $|\hat{\omega}|^2 = 4\pi^2 |k|^2 |\hat{u}|^2$.

Proof. Splitting $\hat{u} = a + ib$, for each real part $\sum_{ij} (k_i a_j + k_j a_i)^2 = 2(|k|^2 |a|^2 + (k \cdot a)^2)$ and $|k \times a|^2 = |k|^2 |a|^2 - (k \cdot a)^2$; both $k \cdot a$ and $k \cdot b$ vanish. □

Theorem 8 (`L2_strain_eq`)**.** *For L^2 fields with these coefficients, $\|S\|_2^2 = \frac{1}{2} \|\omega\|_2^2 \leq \|\omega\|_2^2$.*

Proof. Parseval on each component, exchange the finite and infinite sums, apply the lemma per mode. □

5 The curl of the energy derivative

$\hat{\psi} = \hat{\omega} / (4\pi^2 |k|^2)$, $H(\hat{\omega}) = |\hat{\omega}|^2 / (8\pi^2 |k|^2)$.

Theorem 9. *For $k \neq 0$, $k \cdot \hat{u} = 0$: $H(\operatorname{curl} \hat{u}) = \frac{1}{2} |\hat{u}|^2$; $H(\omega + \varepsilon \eta) = H(\omega) + \varepsilon \operatorname{Re} \langle \hat{\psi}, \eta \rangle + \varepsilon^2 H(\eta)$, so $\delta H / \delta \omega = \hat{\psi}$; and $2\pi i \, k \times \hat{\psi} = \hat{u}$, i.e.* $\operatorname{curl}(\delta H / \delta \omega) = u$.

Proof. $k \times (k \times \hat{u}) = k(k \cdot \hat{u}) - |k|^2 \hat{u} = -|k|^2 \hat{u}$, so $(2\pi i)^2 k \times (k \times \hat{u}) / (4\pi^2 |k|^2) = \hat{u}$. The expansion is the polarisation of a real quadratic form. □

Commentary. The original degeneracy argument read: "2. $\delta H / \delta \omega = u =$ Biot–Savart ω. 3. By definition of Π_u we have $\Pi_u(u) = 0$" (*Tether_May31.lean*, author's unpublished draft), and the May drafts wrote "$\delta H / \delta \mu = u$ (Biot–Savart recovery, Coulomb gauge)" (*May 16 edits*). Formalising it exposed

the slip: as a functional of vorticity, the derivative of the kinetic energy is the stream function ψ, not u, and the ψ-slot tether failed energy degeneracy for that reason (the failure is shown in Lean on an explicit two-mode shear in the public repository `frohmanian-symplectic-tether-regularity`, commit `a52c6af`). The repair was to put the curl into the tether slot, so that what it needs is $\mathrm{curl}(\delta H/\delta\omega) = u$. In the tether build that identity was a hypothesis, because the derivative there is abstract. In Fourier variables it is the theorem above: the earlier statement becomes true once the curl is put back in.

6 Galerkin truncation (finite-mode results)

On the mode box $|k_i| \le N$ the truncated system is a finite-dimensional ODE. Kernel-checked:

- `sobolevProductEstimate`: for $s > 3/2$, $\|fg\|_{H^s} \lesssim \|f\|_{H^s}\|g\|_{H^s}$ with an explicit constant; `summable_fconv_inner`: the convolution series converges absolutely for $f, g \in H^s$, $s \ge 0$.
- `galerkinNS_local_existence` (Picard–Lindelöf) and `galerkinNS_structure_Icc`: Hermitian, divergence-free data stay so on the existence interval.
- `galerkinTransportBound_of_gt`: for $s > 5/2$, $-\mathrm{Re}\langle u, P_N(u\cdot\nabla u)\rangle_{H^s} \le C\|u\|^3_{H^s}$ uniformly in N; `galerkin_uniform_bound_of_gt`: $\|u_N(t)\|^2_{H^s} \le \|a\|^2_{H^s}+1$ on $[0, \texttt{barrierTime}]$, uniformly in N, for Hermitian divergence-free Galerkin solutions.
- `galerkinTameTransportBound_of_gt` and `galerkin_hs_propagation`: if $\|u_N\|_{H^{s_0}} \le A$ ($s_0 > 5/2$) on $[0,T]$ then $\|u_N(t)\|^2_{H^s} \le \|u_N(0)\|^2_{H^s} e^{CAt}$ for $s \ge 2$.
- `galerkinNS_continuation_of_hsBound`: an H^s a priori bound on $[0,T]$ gives a Galerkin solution on all of $[0,T]$.

7 Build status, conditional inputs and scope

Proved. `lake build` of all 14 modules of `TorusFourierNS` succeeds; the axiom report (`AxiomReport.lean`, listed in `AXIOMS.md`) finds 250 theorems, none using `sorryAx`, and no axiom beyond `propext`, `Classical.choice`, `Quot.sound`. This is a machine check of the Lean statements, not peer review.

Conditional. Results whose hypotheses are not discharged here: the a priori bounds in `galerkin_hs_propagation` and `galerkinNS_continuation_of_hsBound`. The named propositions `SobolevCommutatorEstimate`, `GalerkinUniformHsBound` (for all times) and `GalerkinCompactnessLimit` are stated but not proved.

Open. The range $3/2 < s \leq 5/2$ of the transport bound; passage from Galerkin solutions to solutions of the Navier–Stokes equations; L^p strain bounds for $p \neq 2$; and global existence or regularity of 3D Navier–Stokes, which this work does not address.